MT201 – Operations Management

Assignment 3 – Group Case Study

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Introduction

Rajarambapu Co-operative Sugar Factory is a cooperative sugar factory founded in 1969 in Maharashtra, India. We are presented with the view of Suraj who is the general manager of the factory. Suraj has proposed a new strategy for the capacity planning process of the company but is facing pushback from his fellow managers who also have familial ties to the managing director. Suraj has been deeply involved with the planning of this new strategy and is eager to experiment and make it work but he feels he and other managers will be blamed if anything goes wrong.

Question 1: What Challenges did Suraj face?

One of the first problems that Suraj mentioned was the staffing issue that they had. During their prime and optimum season of the overhaul, they could have employed extra staff for the season to do part-time work so they can fully and properly optimise the amount of sugar they produce for the business.

The labour costs that Suraj tells are also a thing for the company. Extra work and pay for the staff. To optimise people's working hours by having 2 shifts of 12 hours a day instead of having 3, 8-hour shifts. This would increase the flow and the movement of the product. By running a 12-hour shift there are no people leaving the workflow 2 times to slow production but only once per 24 hours.

The labour market is another problem that Suraj and his team face. There are not enough “skilled workers'' available for them to work with. The more skill they have the more quality

control and more of the production flow will be. They don't have a chance to increase wages because of the average skill of the workers.

Another problem the Suraj and team have is the information and the sources they have to get them. They are speaking with the bottom line of the company and reporting and trying to convince the higher-ups so that the company can be more efficient. There is not enough evidence that they have yet to prove that they can be better. The HR team is looking at the 12-hour shifts and thinking of the workload that each of their employees has. They are also looking for part-time workers who can also do work if they have the skills and capabilities.

By having a non-uniform workload based on the seasonal sales and production the company will have to keep going through the staff that is available. There won't be any need for the company to have lots of staff in its off-season but they will need more in the January and February time as they are producing the most in these two months. By not having fixed and skilled workers they are not efficient in the business to make the maximum amount of profit they should be getting.

Question 2: Outline the basics of APP, and state the assumptions noted throughout the case.

Aggregate Production Planning (APP) is a medium-term capacity planning process that identifies the lowest cost, workforce and production planning required to meet the demands of customers (Cheraghalikhani et al., 2019). It tries to create desirable production, inventory and employment levels to fulfil the total demand for all products that have the same limited resources (Cheraghalikhani et al., 2019). The three strategies of APP that Suraj mentioned implementing are chase, level and subcontracting, he believes they will optimise production. With a chase strategy, the amount of the workforce is altered to meet or chase demand (Jamalnia *et al.*, 2017). With a level strategy, the amount of time output will be constant any changes in demand will then be met and dealt with using overtime, temporary workers, subcontracting and many other demand options (Jamalnia *et al.*, 2017).

Within the chase strategy, while it is extremely popular among operations managers its extreme changes of hiring, laying off staff and in Suraj's case hiring part-time workers and only retaining 25% of labourers can lead to a huge loss in productivity (Jamalnia *et al.*, 2017). Persistent laying off can cause instability and disorientation within the workplace which can lead to this loss in productivity and motivation among the employees (Jamalnia *et al.*, 2017). We can see this frustration from the employees in the case study, Anoop and Laxmi express their disdain towards this idea Suraj proposes. Anoop even points out that due to their labour market being so competitive once they lay off employees they could lose them permanently and would not return due to the constant scrutiny that comes with the chase strategy.

Suraj explains that in the level production strategy regardless of demand the average monthly quantity will be produced every month. While this can help keep production levels stable, keep employees and increase efficiency, Anoop points out it can add to the total cost. The level production strategy would be more applicable when demand for the product or service is valued highly by customers and will wait for their demand to be satisfied (Armistead and Clark, 1994). For these reasons, I also don't believe the level production strategy is suitable for sugar production. The sugar market can take seasonal patterns for example holidays might see a change in demand for sugar, making your production able to fluctuate would be much more practical for this company.

Finally, Suraj tells us subcontracting will make an external third party produce a portion for them which would lead to huge efficiency and eliminate the need for a warehouse which could cut costs. Jagdish explains this however could risk the quality of the sugar. If the quality of the sugar decreases it would be a deterrent for consumers to buy their product again. However due to demand being unpredictable and customers' preferences being able to charge subcontracting can then impose higher costs and an increase in backorders which could lead to customer dissatisfaction (Jamalnia *et al.*, 2019).

Question 3: What should Suraj recommend to minimise costs and meet the demand for production? Should he recommend continuing with three 8 hour shifts? Or should he recommend implementing two 12 hour shifts?

There are many factors to consider when a huge change in production for a factory is suggested. A comprehensive analysis of each option must be greatly explored before reaching a final decision. Initially, we would have been keen to recommend introducing the new strategy of two 12 hour shifts as it has never been trialled within the company. It is unknown whether it would be beneficial to minimise costs and meet the demand for customers.

The company has always worked in a three 8 hour shift rotation, and yes that has benefited the company and proved to be a success but a new method could be more optimal in efficiency. When a decision would directly impact the company's employees it is important for it not to be made in isolation of the staff. As mentioned in the case study, their current labour market is extremely competitive and the company do not want to end up in a situation where their poor decision making has led to a loss in their skilled workforce. Whatever decision is taken it must not jeopardise the potential risk of losing their skilled work force. Implemented a 12hour shift rotation would mean: extended days off for staff, decreases in shift changes increasing productivity, operations are running smoother due to decrease in interruptions from changeover, decrease cost in overtime hours. However, long shifts can cause fatigue increasing risk of error, productivity, efficiency, and safety can become comprised near the end of a shift.

Whereas remaining with the 8hour shift rotation which is working perfectly well for the business enables: ability for production to operate continuously, greater flexibility for employees- better work-life balance, decreased possibility of the need for overtime costs, variety for employees, increased productivity throughout shift, while challenges can be caused with communication due to the three shift rotation. A final decision cannot be discussed without statistics:

Labour Costs for three 8 hour shifts: 80 x 3 = 240 PER WEEK --- 12480 ANNUAL 1 EMPLOYEE

Labour Costs for two 12 hour shifts: 180 x 2 = 360 PER WEEK -- 18720 ANNUAL 1 EMPLOYEE

Production Efficiency three 8 hour shift: 100% per shift – 100% Efficiency

Production Efficiency two 12 hour shift: 100% for 8hours then 50% for remaining 4hours per shift --- 100/150 x 100 = 66.6% Efficiency

After analysis it would not be in the company's best interest to change the current shift cycle. Maximising profit, minimising cost and achieving optimal production efficiency is the main goal of the company and a change to shift rotation would not only create a large increase in wage costs but also would decrease overall efficiency for Rajarambapu. Suraj should recommend to continue with the three 8 hour shifts to best benefit the future of the company.

Question 4: Considering efficiencies and the cost of labour , calculate the optimal number of regular and part-time labourers to be employed?

From the data we see that regular workers are 100% efficient and part time labourers are 70% efficient. We also see that this directly reflects on the pay differences of 10/hr and 7/hr respectively (Hirsch, 2005). We must consider that the information given indicates that during the maintenance period hiring and firing costs are negligible, while during the production period these acts incur heavy costs. In order to find the optimal number of each type of labourer for the next period (maintenance), we must also consider the effect of the starting number of each type of labourer for the production period. From question 3 we already have discovered that employing labourers to take part in three 8 hour shifts a week is more beneficial that two 12 hour shifts a week, so we will not consider this variable.

Below is data of Labour Hours required monthly based on historical data.

Month	2012	2013	2014	2015	2016	Labour hours
Sep						21932
Oct	26500	26250	26850	26400	26150	34359
Nov	21050	21000	21450	21300	21650	27677
Dec	23000	24500	28500	24500	25700	32812
Jan	50000	52300	54100	52900	53700	68380
Feb	41700	40300	41700	42300	45600	55016
Mar	31500	32800	30700	31900	32600	41470

Below is data on variations of Regular and part time workers when going into the production cycle from the maintenance cycle.

Month	25%Regular	75%Part time	50%Regular	50%Part time	75%Regular	25%Part time
Sep	29	85	52	52	72	24
Oct	45	134	82	81	112	38
Nov	36	108	66	65	90	30
Dec	43	128	78	77	107	36
Jan	90	266	163	161	223	75
Feb	72	214	131	130	180	60
Mar	54	161	99	98	135	45

Below is the calculated costs when using different amounts of regular and part time labourers. This includes wages and, hiring and firing costs.

	25%Regular	75%Part time	50%Regular	50%Part time	75%Regular	25
Hiring and Firing Costs	160028	474808	290512	288402	398838	
Total	**634835**		**578914**		**564893**	
	25%Regular	75%Part time	50%Regular	50%Part time	75%Regular	25
Wages	844071	1227149	1532313	745379	2103683	
Total	2071219		2277692		2449103	
Total Costs	**2706055**		**2856605**		**3013996**	

Using these calculations, we can see that there is a correlation between cost saving and have a lower number of regular labourers to begin with. Although these calculations have only

been done for the chase strategy, we can safely assume that the same results would appear if we did them for the level and subcontracting strategies (Jamalnia et al., 2017). The level strategy as, the average value would be 43,286 labour hours required, meaning that the beginning number of labourers would not differ much. The subcontracting strategy as, the lowest number of labourers needed will be chosen and this would decrease spending even more as hiring and firing costs would be even less.

We know that the lowest number of regular labourers must equal to 25% of the total value (140 labourers), so we first can work out the optimum number of regular labourers. This can be calculated using the following formula.

Optimal regular labourers	=	Total Labourers	x	25%

Using this formula we can assume that 35 regular labourers is optimal.

We can now consider that there is 22 weeks in the maintenance period, where all labourers work three 8 hour shifts a week. In this case study, we assumed that part time and regular workers work the same number of hours as there is no data showing otherwise.

35 labourers X 1 X 22w X 3d X 8h = 18480

18480 ÷22w÷3d÷8h ÷0.7 = 247 labourers

We started with the optimal number of regular labourers and multiplied it by the efficiency and, weeks, days and hours of work to get the total number of labour hours provided by regular labourers. We then divided this number by the values associated with part time labourers to get the optimal number of part time labourers.

In conclusion, the optimal number of regular labourers is 35, and the optimal number of part time labourers is 247.

Question 5: What are the business implications of Suraj's decisions?

The business decisions that Suraj made are considering changing the shift hours, seasonal staffing, and optimising production. The decision of changing the shift hours is changing them to two 12 hour shifts a day from three 8 hour shifts a day. The implications of this decision being made are the possibility of a more motivated workforce as the shifts are 8 hours are shorter, employees will have better attention spans as they are in for less time,

and employees are less likely to get as tired as quickly (Struckman, no date). The disadvantages of the 3 by 8 hour shift a day is that there is more of a staff changeover which can result in miscommunication and disruptions in work.

If Suraj changes to the two 12 hour shifts, this will benefit the company as there will be fewer staff change overs resulting in less work errors, it can also lead to less absenteeism in the workplace because employees are less likely to lose a 12-hour shift than an 8-hour shift. The disadvantage of the 12-hour shift is that employees are more likely to get tired. (*Pros and Cons of 12 Hour Shifts - What's Best for Your Business*, no date).

The next business decision is seasonal staffing which Suraj wants to meet staff requirements to match with the trends in the overhauling season. This will allow the business to have the correct number of staff at a time and when needed.

The next decision Is to optimise production. They want to introduce an aggregate production planning, APP, strategies namely, chase, level, and subcontracting. These strategies aim to make the most out of the sugar production by looking at the different factors that affect it such as staff and demand. These decisions will impact the business as they will reduce the cost for the business because the APP strategies aim to keep staff requirements in line with trends along with inventory, storage, and expenses. This APP strategy will impact the business as it will also ensure that the correct levels of stock, demand is met during the production and by the workforce, warehouse quantities are managed and the subcontracting.

Conclusion

Overall, Suraj is facing a lot of adversity and doubt when it comes to dealing with his superiors in the company, as he believes that he has found the optimal way to run the business to produce more efficiency from workers while solving inventory cost problems. Suraj along with his fellow managers have formulated their version of APP so that workers are more efficient in the hours they work by cutting down on long shifts which will ensure employees do not start to become negligent and reduce the quality of the end product. Suraj has also solved an inventory problem by reducing the amount of inventory held and which costs the company as it has to be paid for to be stored in a warehouse. We believe that Suraj has the right idea and has found a more effective way to staff and operate his company that will reduce costs and increase efficiency.

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